Mortgage Prepayment Risk Modeling

KM & Cox · ML & Deep Learning · Competing Risks · Neural Survival · Scenario Analysis

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IR and Credit · MTH 9877 · Baruch College, Spring 2026

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Project Scope

Goal. Build a survival-analysis pipeline for mortgage prepayment on Freddie Mac single-family data, comparing classical, ML, and deep-learning approaches.

What we implemented:

- **Part A.** Kaplan–Meier survival on the full 31.7M-loan population, stratified by FICO/LTV/vintage/purpose
- **Part B.** Cox proportional hazards (origination + vintage-macro covariates), PH diagnostics
- **Part C.** ML models (LogReg, RF, LightGBM) for horizon-specific prepayment prediction; Cox baseline
- **Part D.** Deep Cox model (DeepSurv) + linear-Cox-via-SGD ablation
- **Part E.** Competing risks (AJ), time-varying Cox (Andersen–Gill), neural survival (LongitudinalDeepHit), scenario analysis

Headline result. Deep Cox: AUC $= 0.729$ at $T=60$ vs 0.685 linear baseline ($\Delta = +0.045$). Gap widens monotonically with horizon.

Data: Freddie Mac Single-Family Loan-Level

Population. 31,688,129 30-year fixed-rate mortgages, originated 1999–2025.

Outcomes (cause-specific framing for prepayment):

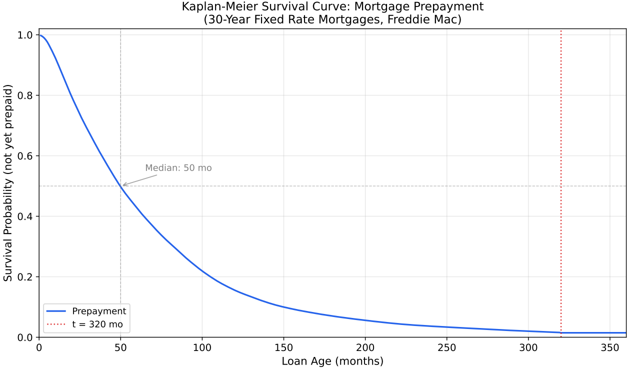
- Prepaid (Zero Balance Code 01): 21,748,774 (68.6%) — the event
- Credit-related termination (ZBC 03, 09): 532,230 (1.7%)
- Other termination (ZBC 02, 15, 16, 96): 265,311 (0.8%)
- Right-censored (still active): 9,141,814 (28.8%)

Survival table. One row per loan with `Duration`, `Event_Prepay`, and 39 origination covariates. Built once via polars, persisted to parquet. Sentinel codes (DTI=999, MI=999, Credit=9999) cleaned to NaN.

Train/test split. By vintage — train: $\leq$ 2014 (full crisis cycle); test: 2015–2019.

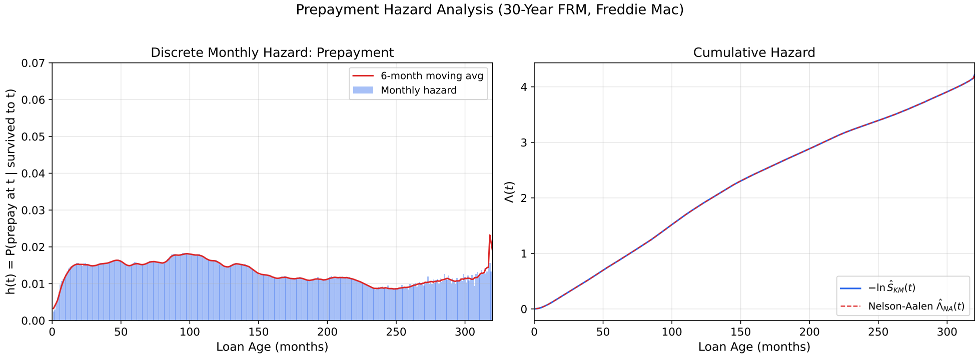
Part A — Kaplan–Meier on the Full Population

No sampling: KM fit on all 31.7M loans.



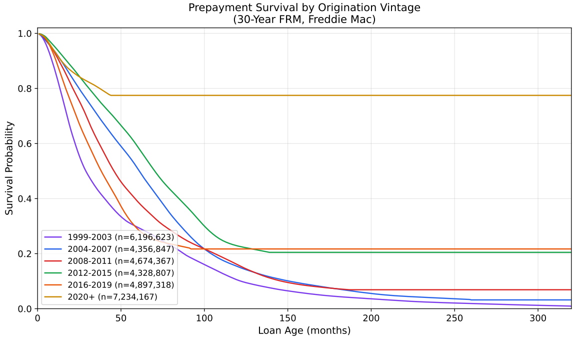
All non-prepay outcomes pooled as right-censored (cause-specific framing): 9.94M censored vs 21.75M events.

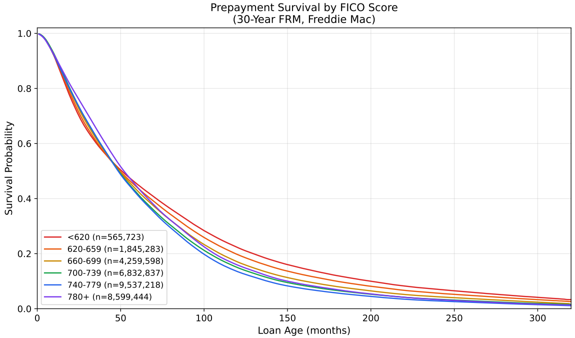
Part A — Hazards



Discrete monthly hazard: ramps from $\approx 1\%$ at $t=6$ to $\approx 1.4\%$ at $t=12$, then plateaus at 1.5–1.7%.
Sanity check: $-\log S_{\mathrm{KM}}$ vs Nelson–Aalen cumulative hazard agree to within numerical precision.

Part A — Stratification by Vintage and FICO





Vintage dominates. 1999–2003: 96% prepay, median 29 mo (post-2000 rate cuts). 2020+: only 16% prepay, median undefined (high-rate environment).

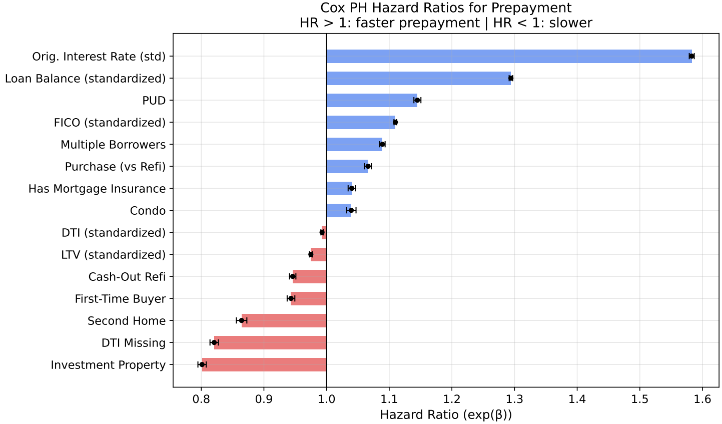
FICO is monotone but modest. Prepay rate ranges from 78% (sub-620) to 65% (780+). Higher-FICO borrowers prepay less in proportion.

Part B — Cox PH with Origination Covariates

Base model. 5% sample (1.58M loans, 1.08M events), 15 origination covariates, lifelines + ℓ_2 penalizer 0.001.
Concordance index: 0.6257.

Most important hazard ratios:

- **Original interest rate:** 1.58
- **Original UPB:** 1.29
- **Investment property:** 0.80
- **DTI missing:** 0.82
- **FICO:** 1.11



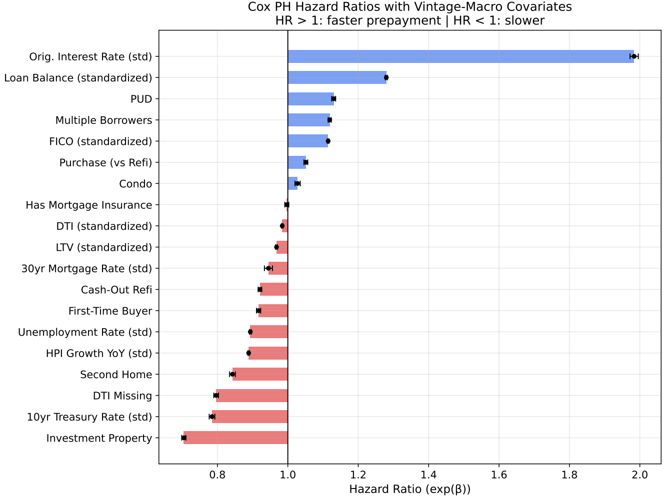
Part B — Vintage-Macro Extension

Macro covariates joined at **origination**. Concordance: 0.6428 ($\Delta C = +0.017$).

Macro hazard ratios:

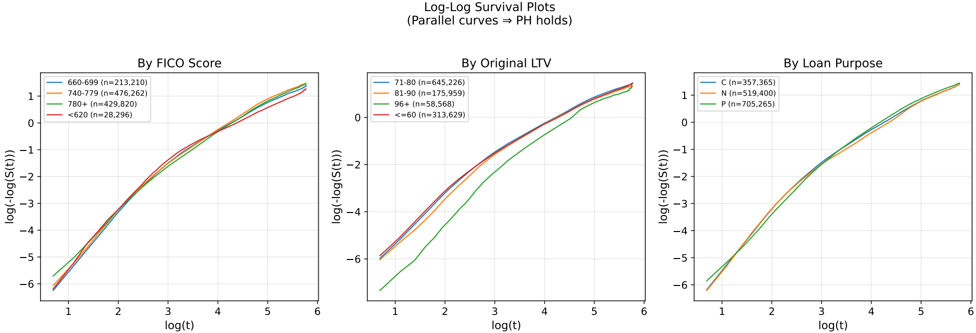
- 10y Treasury: 0.78
- HPI (seasonally adj.): 0.89
- Unemployment: 0.89
- 30y mortgage rate: 0.94

Caveat: vintage-macro covariates capture the macro state *when originated*, not time-varying through the loan's life.



Part B — Proportional Hazards Diagnostic

Visual PH check: log-log Kaplan–Meier curves by FICO, LTV, and loan purpose.



- $\log[-\log S(t)]$ vs $\log(t+1)$; under PH, curves should be parallel
- Strata bend and converge/diverge over time — covariate effects are time-varying, motivating ML/deep models

Part C — ML Models for Horizon-Specific Prediction

Reframing. For each horizon $T \in \{12, 24, 36, 60\}$:

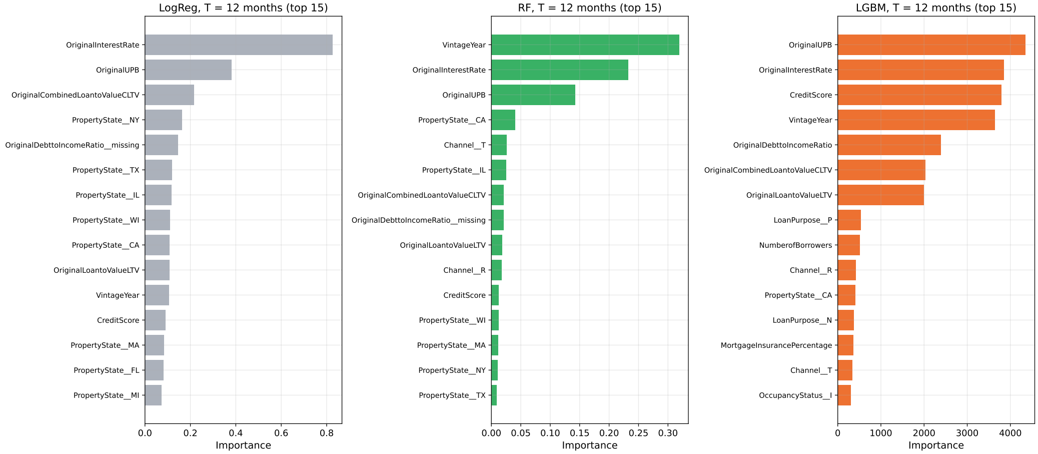
$$P(\text{prepaid by } T \mid X) \quad \text{as a binary classification.}$$

Setup: 10% sample, 1.85M train / 597k test, 68 features after one-hot. Same canonical split as Part D.
Four models:

- Logistic Regression (ℓ_2, internal scaler)
- Random Forest (200 trees)
- LightGBM (with held-out-validation early stopping)
- Cox baseline (lifelines, predict $1 - S(T \mid X)$)

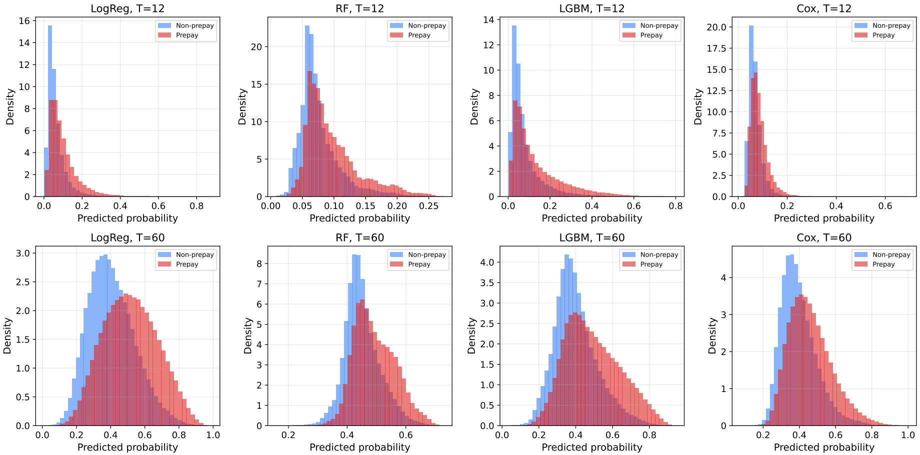
Evaluated: AUC, Brier, log-loss, accuracy — per model, per horizon, stratified by vintage and FICO.

Part C — Feature Importance (T=12)



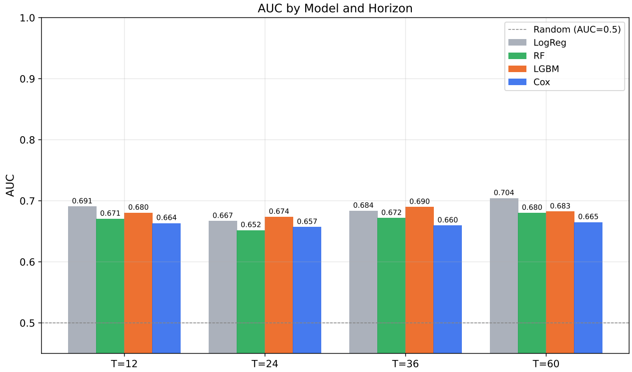
Top 15 predictors for each ML model at the T=12 horizon.

Part C — Score Distributions



Predicted prepayment probabilities at T=12 and T=60, by model and realized prepayment status.

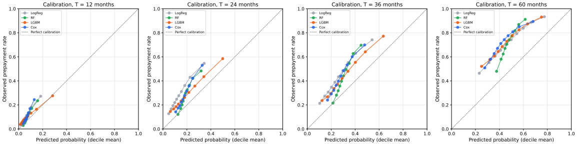
Part C — Out-of-Sample AUC



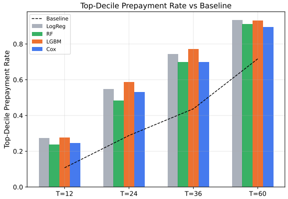
Model	T=12	T=24	T=36	T=60
Cox	0.66	0.66	0.66	0.67
RF	0.67	0.65	0.67	0.68
LGBM	0.68	0.67	0.69	0.68
LogReg	0.69	0.67	0.68	0.70

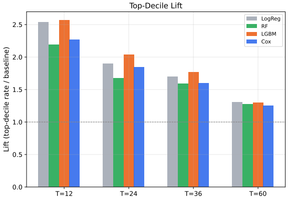
Observations:

- All four models in the 0.65–0.70 band
- LogReg competitive, even best at T=12 and T=60
- Marginal effects largely linear-additive



Calibration is reasonable for all models.





Top-decile lift. At T=12 the top decile catches 27.6% prepayments (LightGBM) vs 10.75% baseline $\rightarrow$ **lift = 2.57**$\times$.

Part D — Deep Cox Architecture and Training

Model: $\lambda(t \mid X) = \lambda_0(t) \cdot \exp\big(f_\theta(X)\big)$ where f_θ is a feed-forward network.

Architecture (DeepSurv-standard):

- 3 hidden layers $\times$ 64 ReLU units
- Batch normalization, dropout 0.1
- 13,184 parameters

Training:

- Adam, $\eta = 10^{-3}$, batch 512
- Up to 50 epochs, ES patience 8
- Breslow baseline post-fit on full train

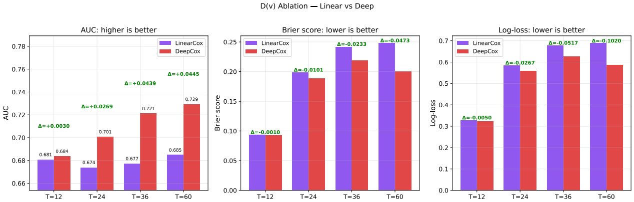
Linear baseline (D(v) ablation):

- Same pycox infrastructure
- `num_nodes=[]` $\rightarrow$ no hidden layers
- 68 parameters; functionally a SGD-fit classical Cox

Why cleanest ablation: same data, optimizer, baseline-hazard estimator — only difference is depth.

Runtime: 17 min Deep + 1 min Linear on CPU.

Part D — Linear vs Deep Ablation (D(v))

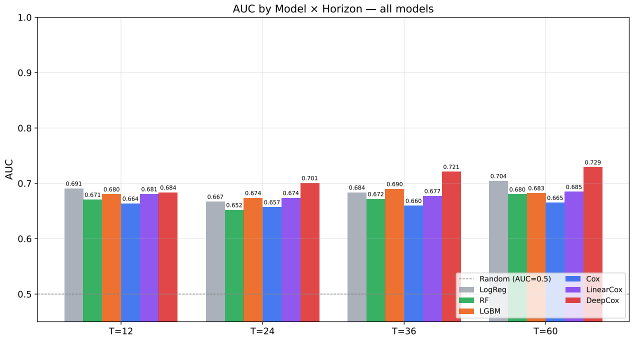


T	Linear	Deep	ΔAUC
12	0.6807	0.6837	+0.003
24	0.6738	0.7007	+0.027
36	0.6772	0.7211	+0.044
60	0.6848	0.7292	+0.045

The gap widens monotonically with horizon.

Sadhwani et al. (2021): nonlinear interactions accumulate over the loan's life.

Part D — Head-to-Head with All Models

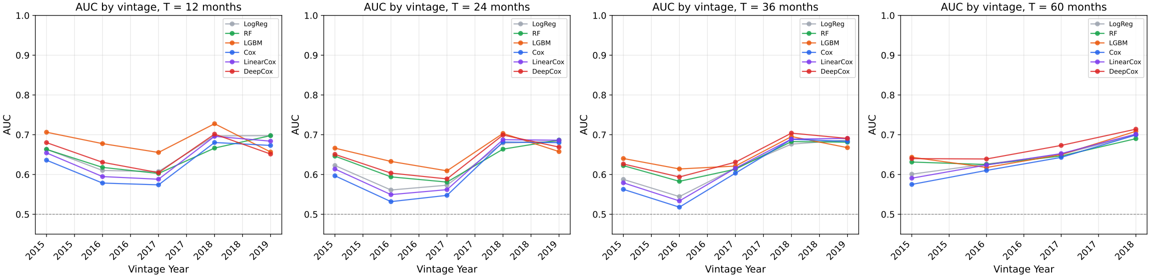


Model	T=12	T=24	T=36	T=60
Cox	.66	.66	.66	.67
RF	.67	.65	.67	.68
LGBM	.68	.67	.69	.68
LinearCox	.68	.67	.68	.68
LogReg	.69	.67	.68	.70
DeepCox	**.68**	**.70**	**.72**	**.73**

DeepCox is best at every horizon $\geq$ 24 months.

At T=12 LogReg edges it — short-horizon prepay dominated by easy cases.

Part D — AUC by Vintage



Why Deep Wins at Long Horizons

Linear assumption: log-hazard is additive in covariate effects.

$$\log \lambda(t \mid X) \;=\; \log \lambda_0(t) + \beta_1 x_1 + \beta_2 x_2 + \cdots$$

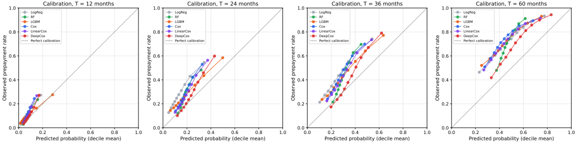
Reality: the refinancing decision at $T{=}36$ depends on FICO, balance, and rate environment *jointly* — non-trivial interactions.

At long horizons, more interaction paths matter. The deep network captures them; the linear model collapses them. Hence the widening gap.

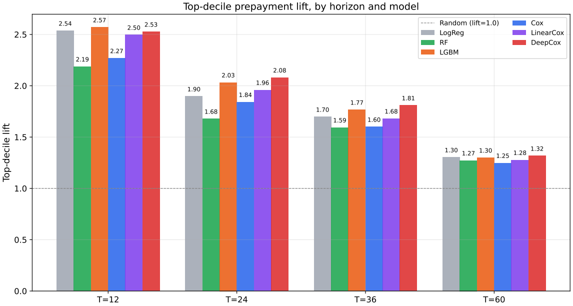
At T=12 only borrowers with very strong, singular incentive have prepaid — easy cases for any model.

Same conclusion as Sadhwani–Giesecke–Sirignano (2021) on the much larger CoreLogic dataset.

Part D — Calibration and Top-Decile Lift



Calibration. DeepCox close to diagonal at all horizons. LinearCox slightly under-confident.



Top-decile lift.

- T=24: DeepCox 2.08× vs LGBM 2.03×
- T=36: DeepCox 1.81× vs LGBM 1.77×

Key Takeaways: Parts A–D

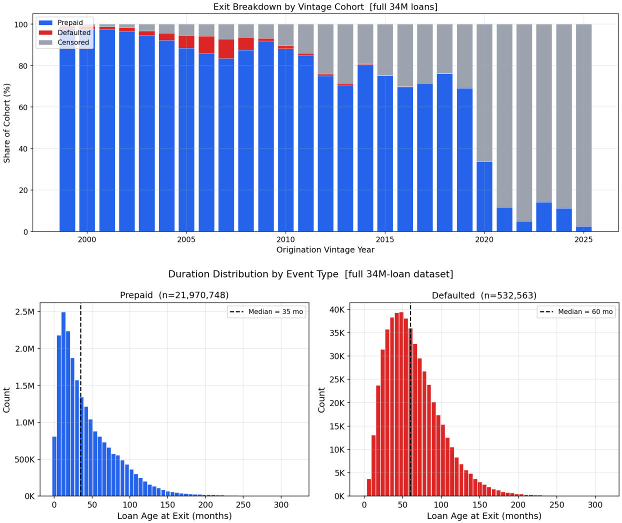
1. **Mortgage prepayment is not a linear-in-covariates process.** Schoenfeld test rejects PH for 12/15 covariates.
2. **Deep Cox AUC = 0.729 at T=60** vs 0.685 linear baseline ($\Delta = +0.045$). Gap grows with horizon.
3. **Tree models don't close the gap.** LightGBM tops out at 0.68–0.69. Interactions are smooth-and-multivariate (good for MLPs) rather than coarse-and-axis-aligned.
4. **Logistic regression is competitive at T=12.** When prepayment concentrates in an obvious refi tail, marginal effects suffice.
5. **Vintage-macro covariates earn their place.** Cox concordance jumps from 0.626 to 0.643 from FRED data alone.

Part E(i)

Competing Risks: Prepayment vs. Default

Kaplan–Meier vs. Aalen–Johansen · Cause-Specific Cox · Fine–Gray

Prepaid and default are competing risks.



Kaplan–Meier vs. Aalen–Johansen.

Kaplan–Meier

$$\hat{F}^{(k)}_{\mathrm{KM}}(t) = 1 - \prod_{t_j \le t} \left(1 - \frac{d_j^{(k)}}{n_j^{\mathrm{KM}}}\right)$$

$d_j^{(k)}$	cause-k exits at t_j
n_j^{KM}	**risk set excluding competing-event loans**

Aalen–Johansen

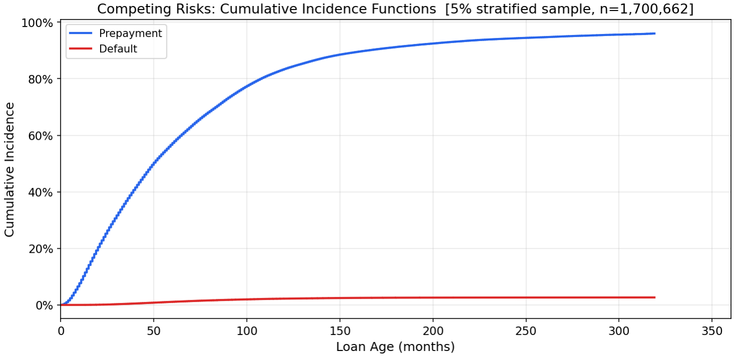
$$\hat{F}_k(t) = \sum_{t_j \le t} \hat{S}(t_j^-) \cdot \frac{d_{kj}}{n_j}$$

d_{kj}	cause-k exits at t_j
n_j	risk set including all exits
$\hat{S}(t_j^-)$	**overall survival just before** t_j

KM: competing exits censored $\Rightarrow$ denominator too small $\Rightarrow$ each increment inflated.

AJ: $\hat{S}(t_j^-)$ shrinks with every exit $\Rightarrow$ $\hat{F}^{(\mathrm{pre})} + \hat{F}^{(\mathrm{def})} + \hat{S} = 1$ always.

Aalen–Johansen CIF: coherent probability partition.



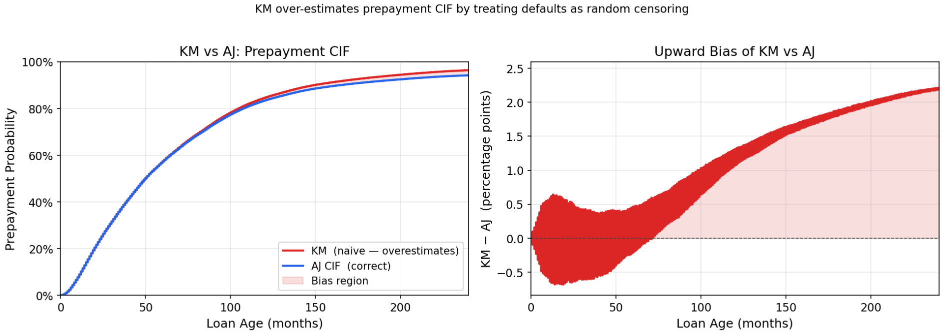
Cumulative Incidence Function

$$F_k(t) = P(\text{exit by } t,\ \text{cause } k)$$

Aalen–Johansen CIF partition

	Prepaid	Default	Active
5 yr	61.4%	1.1%	37.6%
10 yr	83.3%	2.2%	14.5%
20 yr	91.8%	3.1%	5.1%

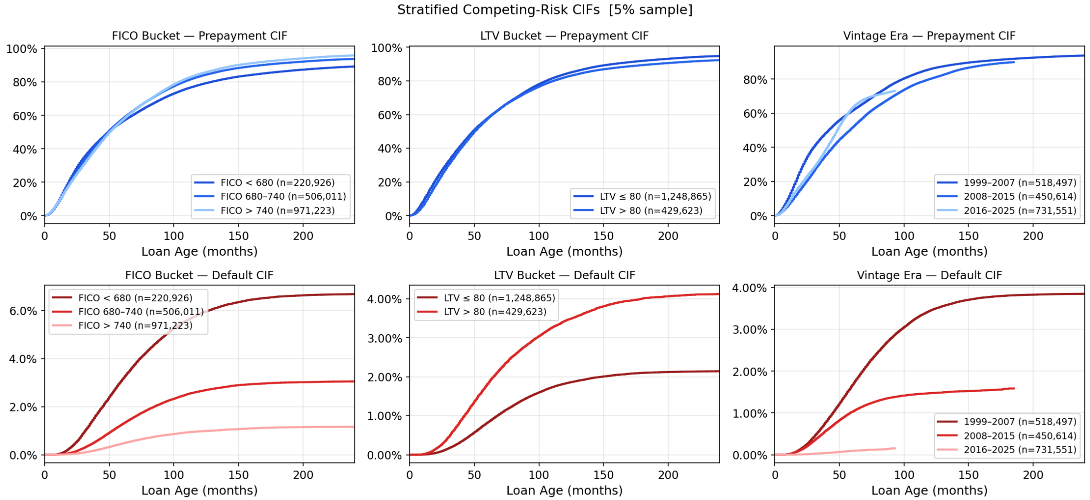
KM overstates prepayment by up to 2.2 pp. over 20 years.



KM overstatement of prepayment probability

Horizon	$\hat{F}_{\mathrm{KM}}$	$\hat{F}_{\mathrm{AJ}}$	Bias
5 yr	61.9%	61.4%	+0.45 pp
10 yr	84.6%	83.3%	+1.31 pp
20 yr	93.4%	91.8%	+2.23 pp

FICO and LTV separate default, not prepayment.



Cause-specific Cox model.

Cause-Specific Cox

$$h_k(t \mid \mathrm{x}) = h_{0k}(t)\, \exp\big(\boldsymbol{\beta}_k^\top \mathrm{x}\big), \quad k \in \{\text{pre}, \text{def}\}$$

$$\ell(\boldsymbol{\beta}_k) = \sum_{i:\, E_i = k} \left[\boldsymbol{\beta}_k^\top \mathrm{x}_i - \log \sum_{j \in \mathcal{R}(t_i)} e^{\boldsymbol{\beta}_k^\top \mathrm{x}_j} \right]$$

$\mathcal{R}(t_i)$: all loans still active just before t_i.

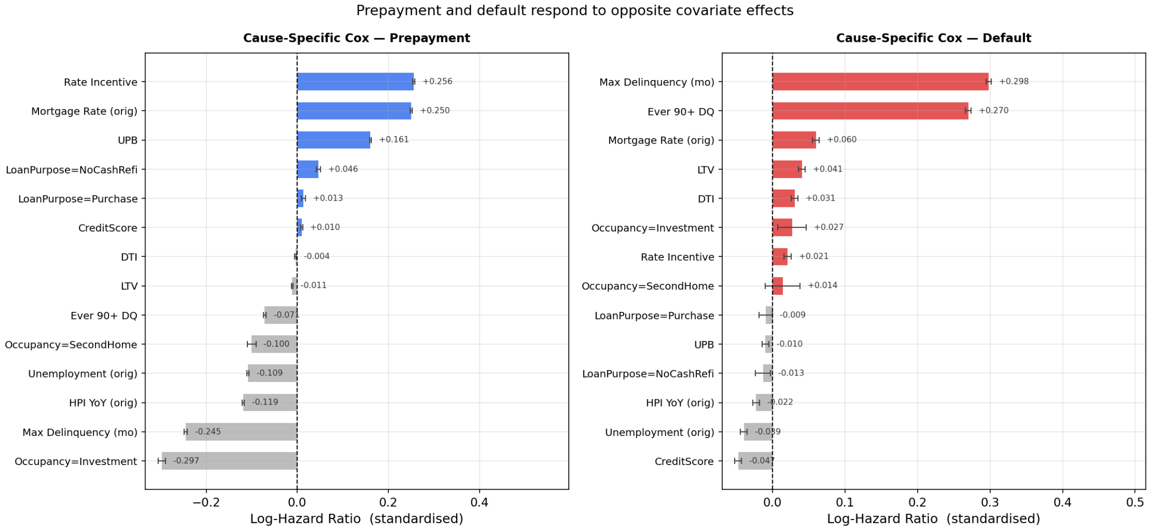
Competing-cause exits are **censored** — they leave $\mathcal{R}(t)$ without contributing to ℓ.

Additional features

Max Delinquency — months past due over loan life

Ever 90+ DQ — binary flag for severe delinquency

Opposite covariate effects by cause.



Four covariates flip sign across causes.

Naive Cox vs. cause-specific coefficients

Covariate	**Naive $\hat{\beta}$**	**Prepay $\hat{\beta}$**	**Default $\hat{\beta}$**	
Rate incentive	$+0.253$	$+0.256$	$+0.021$	(pure prepay)
UPB	$+0.161$	$+0.161$	-0.010	(pure prepay)
FICO	$+0.016$	$+0.010$	-0.047	**opposite**
LTV	-0.007	-0.011	$+0.041$	**opposite**
Max Delinquency	-0.123	-0.245	$+0.298$	**opposite**
Ever 90+ DQ	-0.035	-0.071	$+0.270$	**opposite**

Each extra month delinquent raises default hazard by 35% and cuts prepayment hazard by 22% — yet the naive coefficient (-0.123) conceals both, collapsing a $+0.543$ spread to near zero.

$e^{+0.298} - 1 = +35\%$ (default); $\quad 1 - e^{-0.245} = +22\%$ (prepay reduction); $\quad$ spread $= 0.298 - (-0.245) = 0.543$.

Key Takeaways: E(i)

1. Default and prepayment are competing risks — a naive Cox model conflates both.
2. **Kaplan–Meier overstates prepayment by up to $+2.2$ pp over 20 years; use Aalen–Johansen.**
3. Vintage era dominates: 2007 cohort defaults at $6\times$ the long-run average.
4. FICO drives default; rate incentive drives prepayment — separate channels.

Part E(ii)

Time-Varying Covariates in Mortgage Survival

Andersen–Gill Cox Extension · Live Rate Incentive · Landmark C-index

Static Cox vs. Andersen–Gill.

Standard Cox (11 features)

One row per loan; all covariates fixed at origination. Cannot track rate cycles, burnout, or equity build after origination.

Andersen–Gill (13 features)

One row per loan-month; $x_i(t)$ updated each period. Extends the same static base with four time-varying features. 100K loans $\times$ $\sim$47 months $=$ 4.68M intervals.

Time-varying features — updated every month

Feature	**What it captures**
`rate_incentive`	$r_{\text{orig}} - r_{\text{mkt}}(t)$: live option value to refinance
`ELTV`	equity from amortisation and HPI gains
`unemployment`	macro labour cycle
`hpi_yoy`	home-price appreciation unlocking cash-out refi

*`unemployment` and `hpi_yoy` also in Standard Cox as origination-month snapshots.

The Andersen–Gill model.

Hazard function (identical in form to standard Cox)

$$h_i(t) = h_0(t)\exp\big(\boldsymbol{\beta}^\top \mathrm{x}_i(t)\big)$$

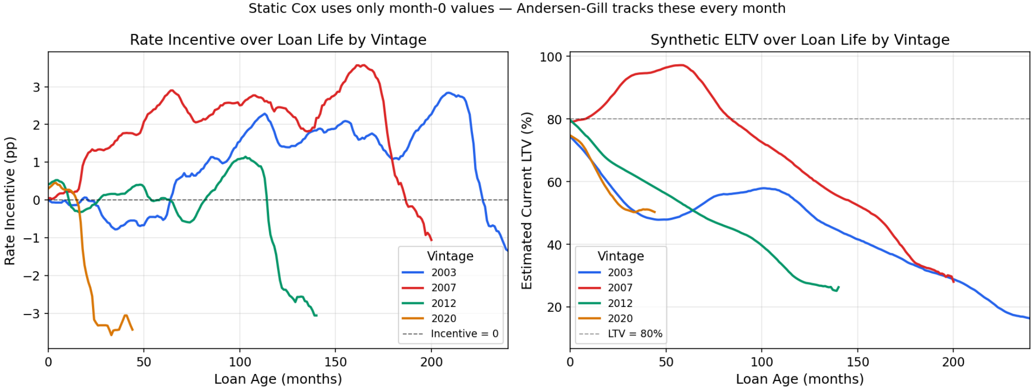
$\mathrm{x}_i(t)$ is the **current** covariate vector, updated each month.

Log partial likelihood

$$\ell(\boldsymbol{\beta}) = \sum_{i:\,\delta_i=1}\left[\boldsymbol{\beta}^\top \mathrm{x}_i(t_i) - \log \sum_{j\in\mathcal{R}(t_i)} \exp\big(\boldsymbol{\beta}^\top \mathrm{x}_j(t_i)\big)\right]$$

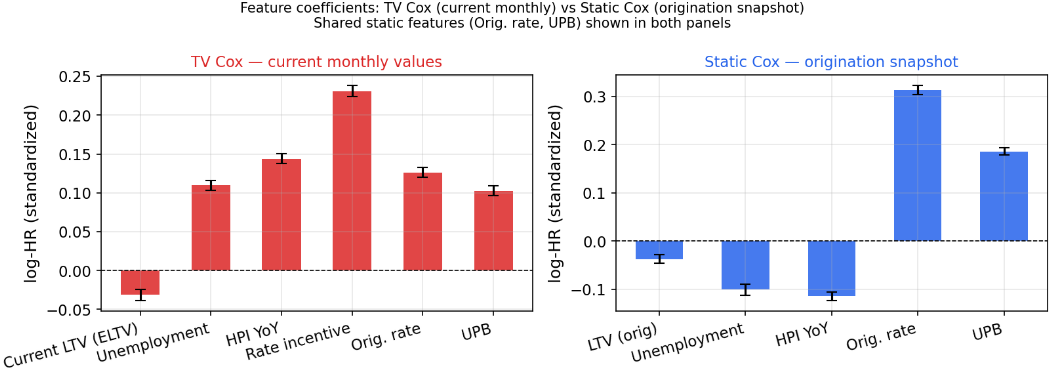
At each event: *given a prepayment at t_i, what is the probability it was loan i rather than any other active loan?*

Time-varying feature trajectories by vintage.



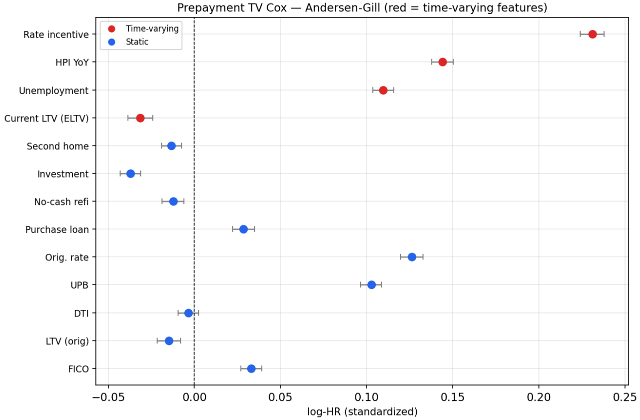
Static Cox uses only month-0 values — Andersen–Gill tracks these every month.

Static vs. TV Cox coefficient comparison.



- `rate_incentive`: dominant positive predictor in TV Cox; near zero at origination, absent from Static Cox
- Macro signals (unemployment, HPI) flip sign — vintage-era snapshots capture a different regime than monthly values

Landmark C-index: TV Cox leads to $T=60$, then reverses.



Landmark C-index

$$C(t^*) = P\big(\hat{h}_i(t^*) > \hat{h}_j(t^*) \mid T_i \leq t^* < T_j\big)$$

Horizon	Static	**TV**	Δ
$T = 12$	0.609	**0.681**	+0.072
$T = 24$	0.627	**0.687**	+0.060
$T = 36$	0.630	**0.675**	+0.045
$T = 60$	0.625	**0.646**	+0.021
$T = 84$	**0.623**	0.605	−0.018
$T = 120$	**0.620**	0.583	−0.037

Key Takeaways: E(ii)

1. Monthly-updated covariates add real information (+1,996 log-likelihood gain).
2. **TV Cox leads at short/medium horizons:** $+0.072$ **at** $T{=}12$**,** $+0.021$ **at** $T{=}60$**.**
3. **TV Cox degrades beyond** $T{\approx}70$ **months — use a landmark super-model.**
4. Rate incentive is the key driver; macro coefficients require careful inspection.

Part E(iii)

Neural Survival Models: LongitudinalDeepHit

Transformer Encoder · Competing-Risk CIFs · Gradient Sensitivity

Why go beyond Cox?

Cox / Andersen-Gill limitations

- Proportional hazards: covariate effect is **constant over time**
- Single scalar score — cannot model the **S-curve shape** of prepayment
- Competing risks require separate model fits; no shared representation
- Cannot exploit the **sequence structure** of monthly covariate history

What a sequence model can do

- Read the full monthly path $[x_1, \ldots, x_T]$ — **burnout is encoded**
- Produce nonlinear, time-varying hazard rates $\lambda_k(t \mid x)$
- Model prepayment and default **jointly** via shared encoder
- Attention weights identify **which months** drive the prediction

Architecture rationale.

Alternatives rejected

- **LSTM** — vanishing gradients at $T{=}120$; attends only to adjacent months; hard to interpret
- **Single-context head** — near-constant hazard curve; gradient flows only through last hidden state
- **Two separate models** — violates independent censoring; loses shared macro signal

Why this combination

- **Causal Transformer** — reaches any past month in one attention step; weights are interpretable
- **Per-timestep head** — $\lambda(t){=}\text{head}(h_t)$: hazard conditioned on full causal history $x_1 \ldots x_t$
- **Joint encoder** — chain product enforces $\text{CIF}_1{+}\text{CIF}_2{+}S{=}1$

LongitudinalDeepHit architecture.

Architecture

Input	$(B,\ T_{\max}{=}120,\ D{=}13)$ padded sequences
Proj.	Linear $D \to d_{\text{model}}{=}64$
PE	Sinusoidal positional encoding on loan age
Enc.	$\times 2$ pre-norm layers, 4 heads, causal mask
Head 1	FC $64 \to 64 \to 1$ per h_t, Sigmoid $\Rightarrow \lambda_1(t)$
Head 2	FC $64 \to 64 \to 1$ per h_t, Sigmoid $\Rightarrow \lambda_2(t)$
CIFs	Competing-risk chain product

Competing-risk CIFs

$$\mathrm{CIF}_k(t) = \sum_{s=1}^{t} \lambda_k(s) \cdot S(s-1)$$
$$S(t) = \prod_{s=1}^{t} (1-\lambda_1(s))(1-\lambda_2(s))$$

Guarantees $\mathrm{CIF}_1 + \mathrm{CIF}_2 + S = 1$ at every t.

Training objective

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \alpha\, \mathcal{L}_{\text{rank}}, \quad \alpha = 0.2$$

NLL: discrete-time cause-specific likelihood
Ranking: soft pairwise concordance (DeepHit-style)

Params	76,290
Data	101K loans, 4.5M rows
Epochs	41 (best ep. 29)
LR	3×10^{-4}, cosine

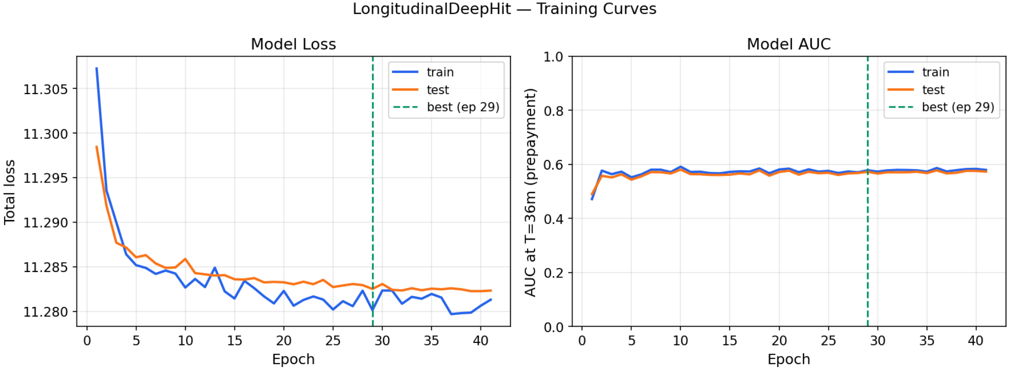
Data treatment

Each loan $\to$ 13-feature monthly sequence, zero-padded to $T_{\max}{=}120$ after exit.

Labels: prepayment $=1$, default $=2$, censored $=0$.

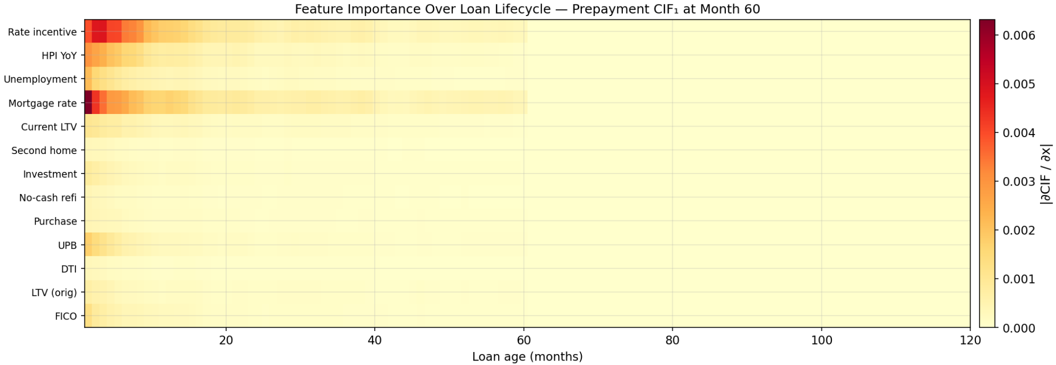
Causal mask; 80/20 train–test split.

Training dynamics: NLL collapse and ranking loss.



- **NLL freezes**: 1.6% event rate means predicting near-zero hazard for everyone already minimises the loss — rare-event gradients too weak to improve further.
- **Cox immune**: partial likelihood only ranks loans at event times, never predicts absolute hazard — no trivial solution under class imbalance.
- **AUC preserved**: ranking loss descends steadily, ordering loans by relative risk despite collapsed hazard values.

Feature importance over the loan lifecycle.



`rate_incentive` dominates, especially in early months; current LTV rises later as equity accumulates.

Model performance vs. DeepCox.

Prepayment CIF_1 — AUC at fixed horizons

Model	T=12	T=24	T=36	T=60
Part D DeepCox†	**0.684**	**0.701**	**0.721**	**0.729**
DeepHit (20K)‡	0.641	0.622	0.572	0.515

Default CIF_2 — AUC at fixed horizons

Model	T=12	T=24	T=36	T=60
Part D DeepCox†	0.684	0.701	**0.721**	**0.729**
DeepHit (20K)‡	**0.687**	**0.708**	0.707	0.726

† 597K canonical ‡ 20K hold-out green = better

Prepayment: DeepCox leads at all horizons (NLL collapse hurts long-horizon calibration).
Default: DeepHit leads at T=12 and T=24 with **6× fewer** training loans.

Key Takeaways: E(iii)

1. Default AUC matches DeepCox with 6× fewer training loans.
2. Feature importance shifts over the loan lifecycle — unique to sequence models.
3. **Limitation: 1.6% event rate causes NLL collapse — AUC is preserved but absolute probabilities are unreliable; fix via focal-loss upweighting.**

Part E(iv)

Scenario Analysis: MBS Prepayment under Rate Shocks

TV Cox Rate Channel · Survival Curves · Monte Carlo MBS Pricing

TV Cox: direct rate channel for scenario analysis.

Why TV Cox (E(ii)) is the primary model

`rate_incentive` $= r_{\text{orig}} - r_{\text{market}}(t)$ is a **direct time-varying covariate**.

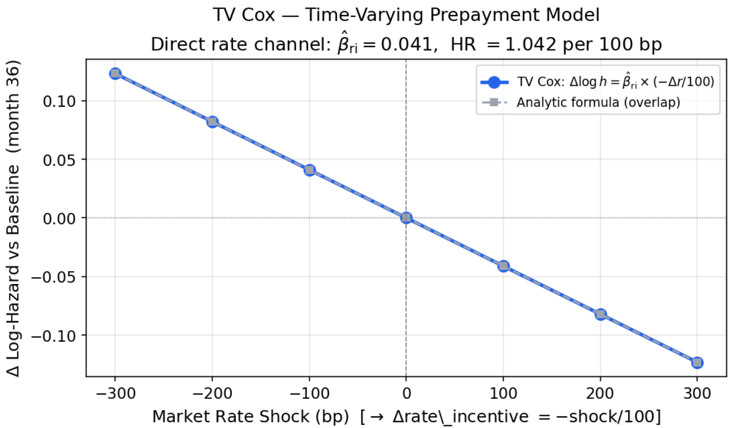
A Δr bp market rate shock immediately shifts the incentive by $-\Delta r/100$ pp, giving an analytic hazard response:

$$\Delta \log h(t) = \hat{\beta}_{\text{ri}} \times \frac{-\Delta r}{100}, \qquad \hat{\beta}_{\text{ri}} = 0.041 \text{ (HR} = 1.042 \text{ per } 100\,\text{bp)}$$

The calibrated baseline hazard $\hat{H}_0(t)$ produces realistic $S(t)$, CPR, WAL, and price.

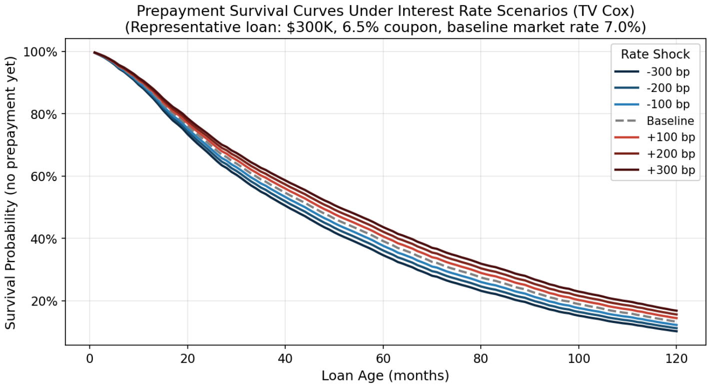
Representative loan: \$300K, 5.0% coupon, baseline market rate 5.3% (median, 1999–2025).
Scenarios: ± 100, ± 200, ± 300 bp — covering the BCBS standard stress range.

Shock sensitivity: TV Cox matches the analytic formula.



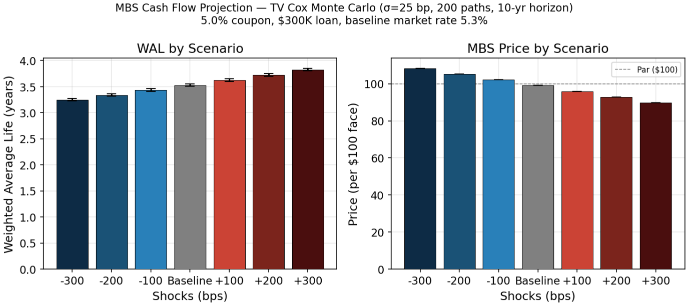
Each 100 bp rate cut increases prepayment hazard by 4.2%; a 300 bp cut raises it by +0.123 in log terms. Model response matches the closed-form prediction exactly — no simulation needed, fully auditable.

Survival curves under seven rate scenarios.



Scenario	10-yr Survival	Median Prepay (mo.)
−300 bp	10.2%	41
−200 bp	11.2%	43
−100 bp	12.2%	45
Baseline	13.3%	46
+100 bp	14.4%	48
+200 bp	15.6%	50
+300 bp	16.8%	51

MBS: Weighted Average Life, price, and negative convexity.



Scenario	WAL (yr)	Price ($)
−300 bp	3.25	108.32
−200 bp	3.34	105.26
−100 bp	3.43	102.17
Baseline	3.53	99.07
+100 bp	3.63	95.96
+200 bp	3.73	92.86
+300 bp	3.83	89.77

Negative convexity: when rates fall, borrowers prepay at par, capping price upside. Gain (−300 bp: $9.25) < loss (+300 bp: $9.30).

MC: 200 paths, σ=25 bp.

Key Takeaways: E(iv)

1. **Static models give the wrong direction — time-varying covariates are required for rate scenario analysis.**
2. **TV Cox has a linear rate channel**: every 100 bp rate cut adds the same +4.2% to prepayment hazard — equal steps in rate produce equal steps in response, making results predictable and fully explainable.
3. **Negative convexity confirmed**: when rates fall, borrowers prepay at par, capping price upside ($9.25 gain $<$ $9.30 loss at $\pm$300 bp).
4. **FICO and LTV effects are real but require more data to surface — rate incentive dominates in this specification.**

Thank you

Questions?

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